Harder Geometrical Problems — problem 12.3.2.88

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The points A_1, B_1, and C_1 are chosen on the sides BC, CA, and AB of $\triangle ABC$, so that AA_1, BB_1, and CC_1 intersect in a point. Let M be the projection of A_1 onto B_1C_1. Prove that MA_1 bisects $\angle BMC$.